The Syntax of the Mandarin *Num-Cl P Num-Cl* Constructions

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Intruduction

Core Data: [They]$_j$ eat [apples]$_i$ [**one by one**]$_{j/i}$.

[他们]$_j$ [**一个接一个地**] $_{j/i}$ 吃[苹果]$_i$。

Puzzle ①: Internal Ambiguity !

Agent or Patient?

(p.c Stiebels:
...dass die Frauen eine nach der anderen Würste gegessen haben.)

EN

[They]$_j$ eat [apples]$_i$ [**one by one**]$_{j/i}$

DE

a. Sie hat [sie]$_j$ gebeten, [PRO]$_j$ [einen nach dem anderen]$_{*j/i}$
she has [them.ACC]$_j$ asked [one.ACC after the other]$_{*j/i}$
[Äpfel]$_i$ zu essen.
[apples.ACC]$_i$ to eat
'She asked [them]$_j$ [PRO]$_j$ to eat [apples]$_i$ [one by one]$_{*j/i}$.'

b. Sie hat [sie]$_j$ gebeten, [PRO]$_j$ [einer nach dem anderen]$_{j/*i}$
she has [them.ACC]$_j$ asked [one.NOM after the other]$_{j/*i}$
[Äpfel]$_i$ zu essen.
[apples.ACC]$_i$ to eat
'She asked [them]$_j$ [PRO]$_j$ to eat [apples]$_i$ [one by one]$_{j/*i}$.'

ZH

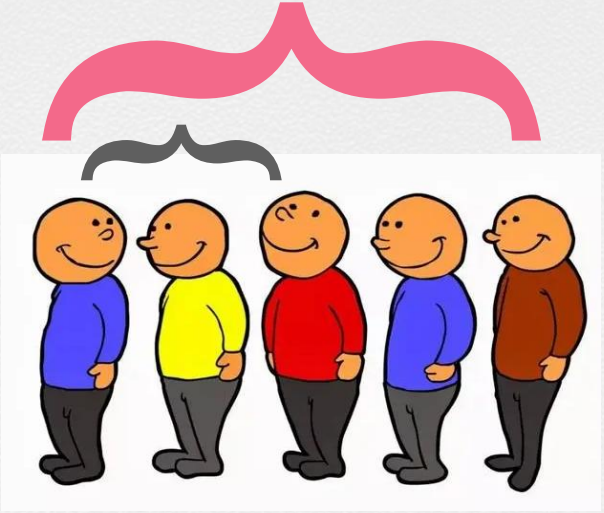
[tamen]$_j$ [yige jie yige]$_{j/i}$ de chi [pingguo]$_i$.
[they-PL]$_j$ [one-Cl following one-Cl]$_{j/i}$ DE eat [apples]$_i$.
'[They]$_j$ eat [apples]$_i$ [one by one]$_{j/i}$.'

Puzzle ②: A Deictic Asymmetry !

Subject-Targeted

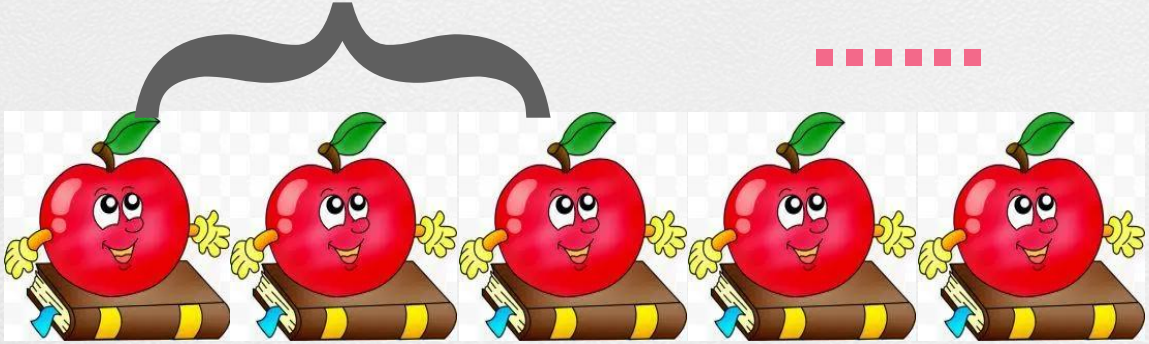
- Every single one
- Just part of them





Object-Targeted

- All apples? **No!**



Puzzle ③: Object-Targeted Reading Disappears !

When we have another quabtificational elements (e.g. ModalP or Negation): → Scopal Relations

a. [tamen]$_j$ bixu/meiyou [yige jie yige]$_{j/i}$ de chi [pingguo]$_i$.

[they-PL]$_j$ must/NEG [one-Cl following one-Cl]$_{j/i}$ DE eat [apples]$_i$.

'[They]$_j$ must/didn't eat [apples]$_i$ [one by one]$_{j/i}$.'

(Modal/Neg > *yige jie yige*: [Subject] √ [Object] √)

b. [tamen]$_j$ [yige jie yige]$_{j/*i}$ de bixu/meiyou chi [pingguo]$_i$.

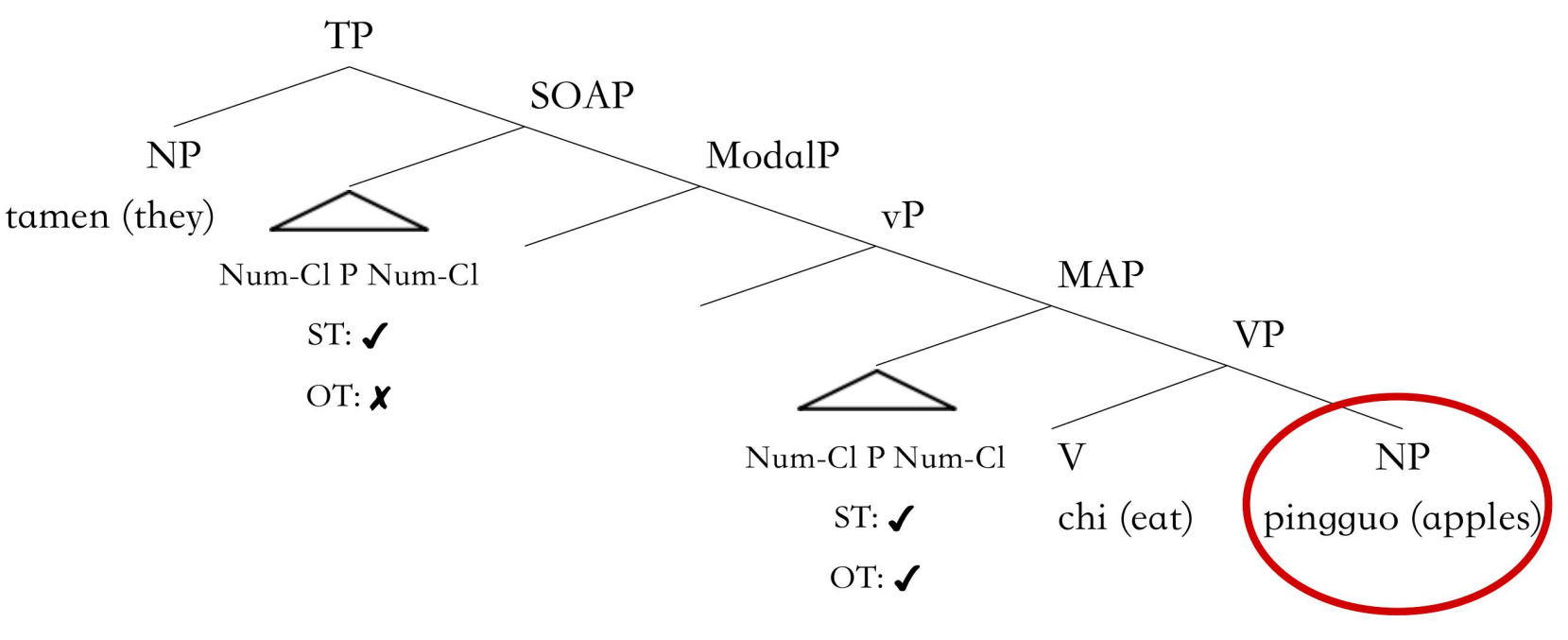
[they-PL]$_j$ [one-Cl following one-Cl]$_{j/*i}$ DE must/NEG eat [apples]$_i$.

'[They]$_j$ must/didn't eat [apples]$_i$ [one by one]$_{j/*i}$.'

(*yige jie yige* > Modal/Neg: [Subject] √, [Object]×)

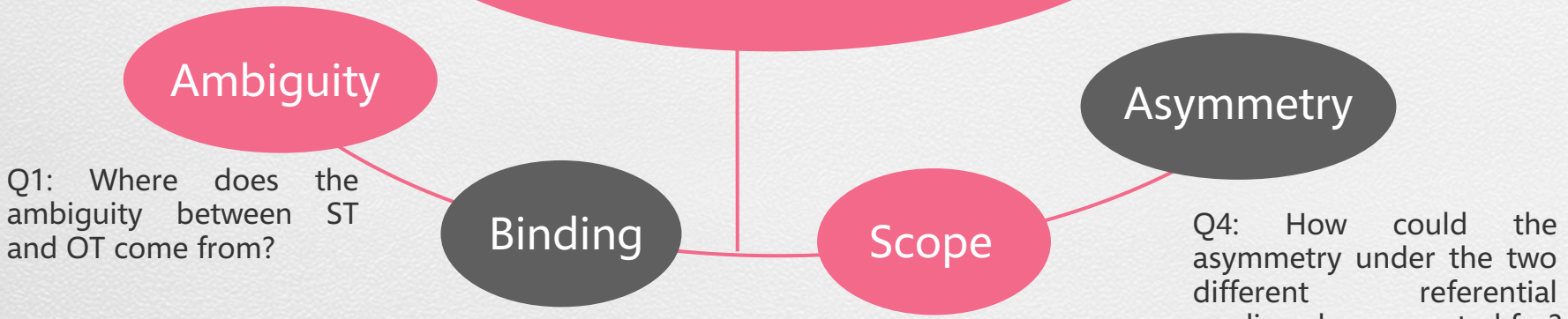
Puzzle ④: Binding but not C-Commanding !

The anaphoric binding requirement must be satisfied, surfacely or deeply : → Reconstruction



Research Questions & Methodology

- Literature Review (comparative)
- Native-speaker Test



Q1: Where does the ambiguity between ST and OT come from?

Q2: How can we establish the binding relationship between the anaphoric phrase *yige jie yige* and its antecedent in object position?

Q3: How do we account for the ungrammaticality of OT in the situation with scopal expressions like modal or Neg?

Q4: How could the asymmetry under the two different referential readings be accounted for?



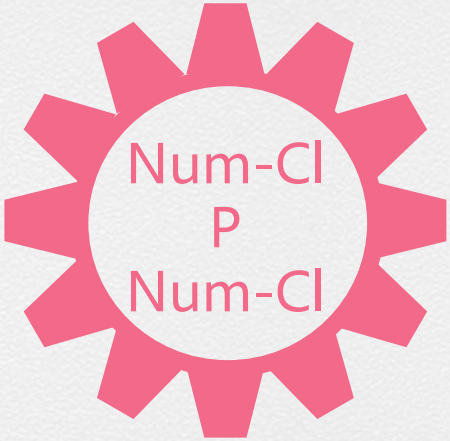
Variations and Distributions

A comparative approach

Comparision between EN and ZH



- Num-by-Num: The students left <u>one by one / two by two</u>.
- N-by-N: They ate the cake <u>piece by piece</u>.



- Num-CL-N P Num-CL-N
- *Num-CL-N P Num-CL
- *Num-CL P Num-CL-N

"*两个接两个"??: "*yi*" (one) has been weakened into a marker of indefiniteness and lost its function as a numeral expression denoting numbers (Yip, 2018).

Num-Cl ***V*** *Num-Cl* or *Num-Cl* ***P*** *Num-Cl*?

Jie（“接”）as Prep.

- Attached with the affix *zhe*（“着”）(a-c), Asp. marker NO!
- Form an independently complete sentence (d-g)? NO!

a. yan = yanzhe
alongside alongside-*zhe*
Prep. Prep.-*zhe*

b. chu = chule
except except-*le*
Prep. Prep.-*le*

c. yige jie-zhe yige
Num-Cl following-*zhe* Num-Cl

d. * nage tongxue jin jie-zhe wo.
Det-Cl student immediately follow-zhe me
Intended: ‘That student immediately follows me.’

e. * nage tongxue bu/mei jie-zhe wo.
Det-Cl student NEG follow-zhe me
Intended: ‘That student doesn’t follow me.’

f. nage tongxue jin ai-zhe wo.
Det-Cl student immediately next-zhe me
‘That student is immediately next to me.’

g. nage tongxue bu/mei ai-zhe wo.
Det-Cl student NEG next-zhe me
‘That student is not next to me.’

Syntactic Positions

a. xueshengmen zenme chi pingguo?
student-Pl how eat apple
'How did the students eat apples?'

b. [xueshengmen]$_j$ [yige jie yige]$_{j/i}$ de chi [pingguo]$_i$.
[student-Pl]$_j$ [one-Cl following one-Cl]$_{j/i}$ DE eat [apples]$_i$
'[Students]$_j$ eat [apples]$_i$ [one by one]$_{j/i}$.' (Adjunct) EN&ZH

c. [yige jie yige]$_{j/*i}$ de [xuesheng]$_j$ chi [pingguo]$_i$.
[one-Cl following one-Cl]$_{j/*i}$ DE [student-Pl]$_j$ eat [apples]$_i$
'One Student after another eats apples.' (Argument-Subject) ZH only

d. [xueshengmen]$_j$ chi [yige jie yige]$_{*j/i}$ de [pingguo]$_i$.
[student-Pl]$_j$ eat [one-Cl following one-Cl]$_{*j/i}$ DE [apples]$_i$
'Students eat one apple after another.' (Argument-Object) ZH only

e. [tamen]$_j$ de [kaoshi]$_i$ (shi) [yige jie yige]$_{*j/i}$.
[they]$_j$ DE [exams]$_i$ (be) [one-Cl following one-Cl]$_{*j/i}$
'They have one exam after another.' (Predicative) ZH only

Distributional Environment (1-2)

Running-time events requires incremental verb, no stative verb.

EN&ZH

a. * The students knew French one by one.

b. * Xueshengmen yige jie yige de hui fayu.
Student-Pl one-Cl following one-Cl DE know French
Intended: 'Every student knew French.'

Covert/overt semantically-plural and non-abstract nominal Argument.

EN&ZH

a. She said good-bye to her [staff]$_j$ [one by one]$_j$.

b. (naxie) pingguo yige jie yige de fulan.
(those) apple one-Cl following one-Cl DE rot
'(Those) apples rotted one by one.'

Mass noun: **only ZH**

a. ta yibei jie yibei de he shui.
he one-glass following one-glass DE drink water
'He drank one glass of water after another.'

b. *John drank water one by one.

Covert target: evidence from **German**

Sie hat [sie]$_j$ gebeten, [PRO]$_j$ [einer nach dem anderen]$_{j/*i}$
she has [them.ACC]$_j$ asked [one.NOM after the other]$_{j/*i}$
[Äpfel]$_i$ zu essen.
[apples.ACC]$_i$ to eat
'She asked [them]$_j$ [PRO]$_j$ to eat [apples]$_i$ [one by one]$_{*j/i}$.'

Distributional Environment (3)

(?) The targeted nominal Argument has to be local in the same clause as the verbal predicate that *one by one* modifies..... (Henderson, 2009)

a. Over those busy decades, as one by one our nestlings fledged and took wing.
(√ fledged one by one)

b. Over those busy decades, as our nestlings fledged and one by one took wing.
(× fledged one by one)

In Chinese, we found this clause-boundedness modification feature only in OT reading, instead of ST reading !

c. [tamen]$_j$ [yige jie yige]$_j$ de chi-le pingguo, you chi-le xiangjiao.
[they]$_j$ [one-Cl following one-Cl]$_j$ DE eat-asp apples, and eat-asp bananas
'They, one person after another, ate apples, and then ate bananas.'
(√ ate apples one person after another; √ ate bananas one person after another)

d. tamen [yige jie yige]$_{i/*k}$ de chi-le [pingguo]$_i$, you chi-le [xiangjiao]$_k$.
they [one-Cl following one-Cl]$_{i/*k}$ DE eat-asp apples, and eat-asp bananas
'They ate one apple after another, and then ate bananas.'
(√ ate one apple after another; × ate one banana after another)



The Internal Structure

Chinese, a typical numeral classifier language

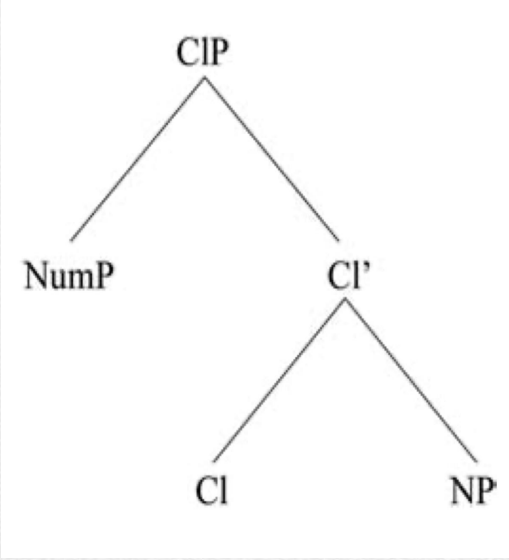
Cl-N Matching Issue

Only when subject and object share the same classifier, ambiguity appears !

ClP Analysis

a. [xuesheng men]$_j$ [yige jie yige]$_{j/*i}$ de kan [wenxian]$_i$.
[student PL]$_j$ [one-Cl following one-Cl]$_{j/*i}$ DE read [paper]$_i$.
'[The students]$_j$ read the [papers]$_i$ [one by one]$_{j/*i}$.'

b. [xuesheng men]$_j$ [yipian jie yipian]$_{*j/i}$ de kan [wenxian]$_i$.
[student PL]$_j$ [one-Cl following one-Cl]$_{*j/i}$ DE read [paper]$_i$.
'[The students]$_j$ read the [papers]$_i$ [one by one]$_{*j/i}$.'



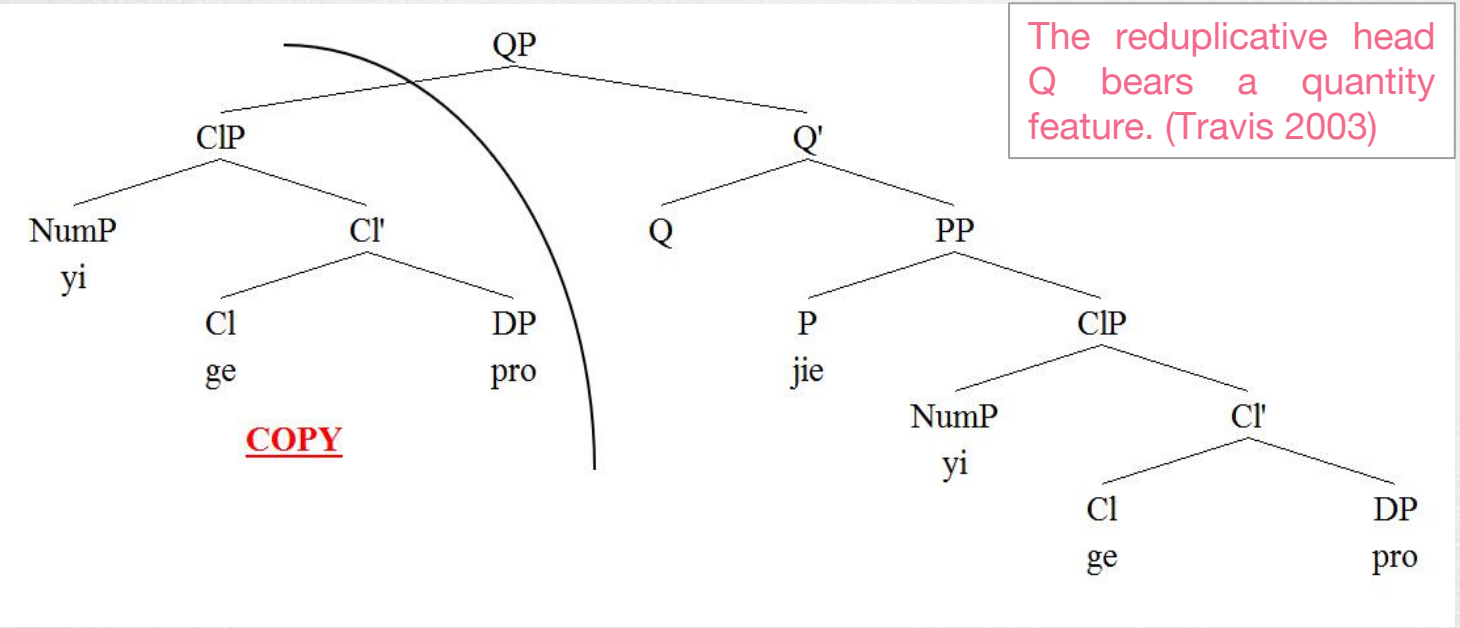
(Zhang, 2009; Pan and An, 2012; Jin, 2019)

Reduplication: the denotation of distributivity

Some data:

a. Student after student visited the professor on Monday. (Germanic Languages)

b. ii pilla-lu renDu renDu kootu-lu-ni cuus-ee-ru. (Telugu)
these kid-Pl 2 2 monkey-Pl-Acc see-Past-3PPl
'These kids each saw 2 monkeys.'



The reduplicative head Q bears a quantity feature. (Travis 2003)

DP

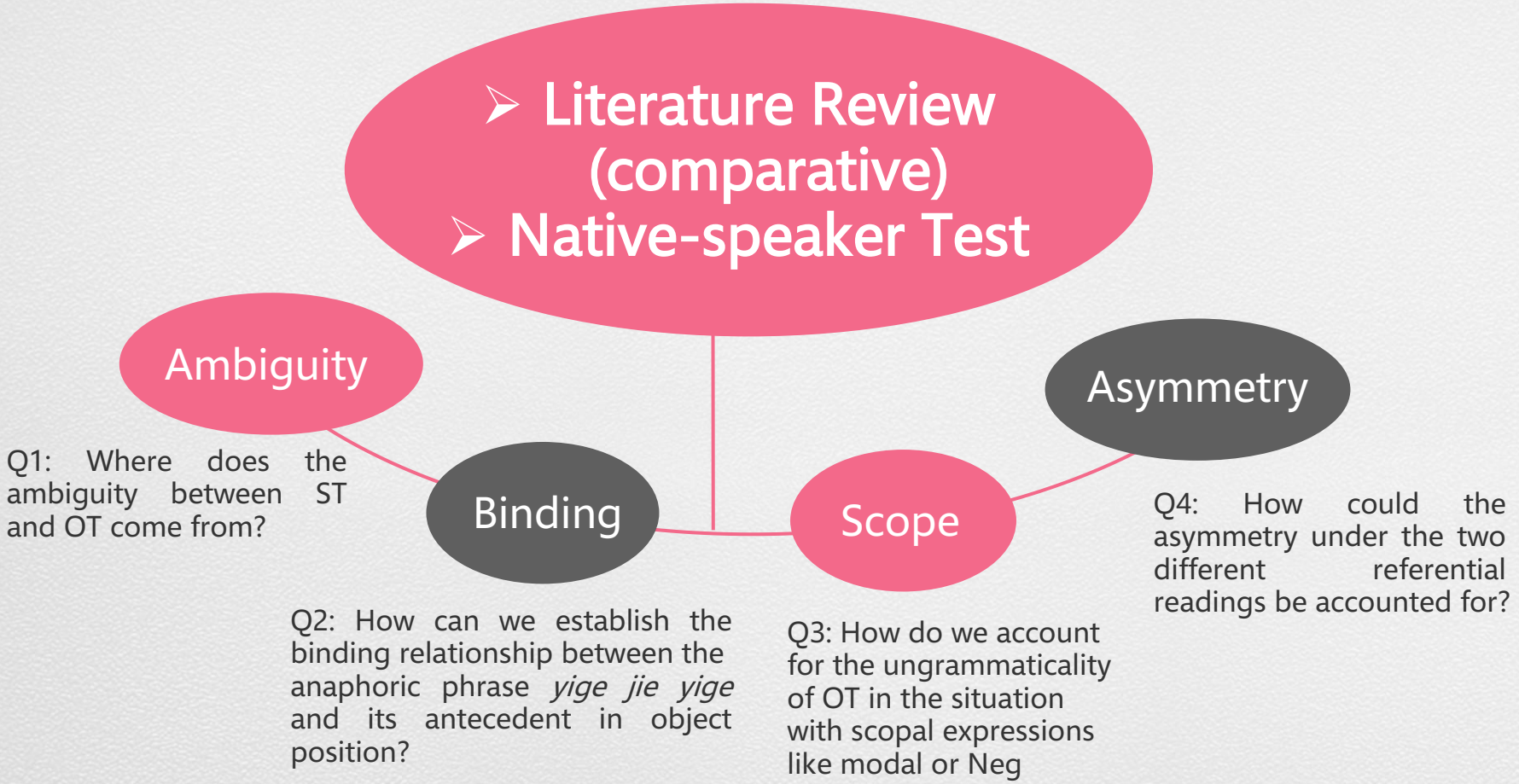
PRO?
DP trace?
DP ellipsis?
NP ellipsis?



Interpretation & External Syntax

Not homonyms, but a fancy story

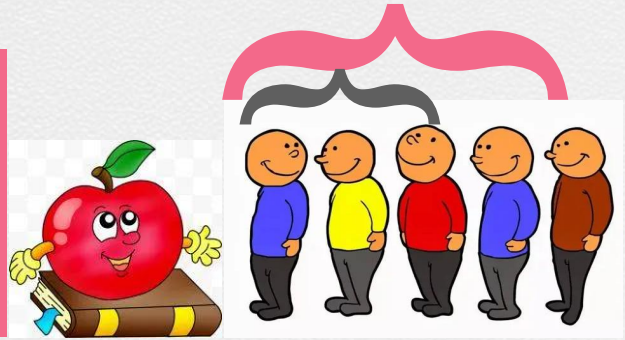
Research Questions & Methodology



Differentiating SOA & MA

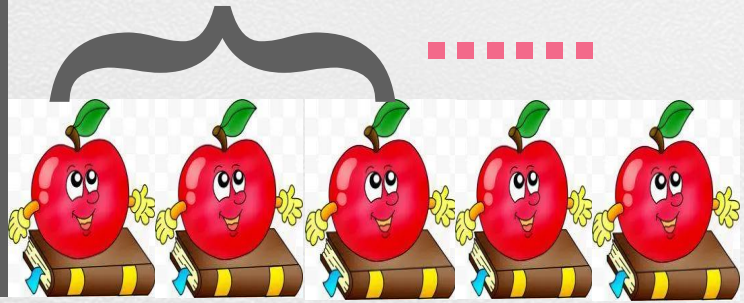
Subject-Targeted

- SOA (Universal quantification)
- MA



Object-Targeted

- Only MA



Subject-Oriented Adv.

emphasizing the quality / feature of the subject

Manner Adv.

event modifier with pure manner interpretation

a. John stupidly danced.

b. Zhangsan congming-*(de) huida-le zhe-ge wenti.
John cleverly-*(DE) answer-asp this-Cl question

(i) 'The manner in which John answered this question was clever.'
(ii) 'It was clever of John to have answered this question.'

(Jackendoff, 1972; McConnell-Ginet, 1982; Ernst, 2002; Kubota, 2015; a.o.)

- A 4-way Ambiguity : (SOA / MA) * (ST / OT) = 2*2 = 4

External Internal

[tamen]$_j$ [yige jie yige]$_{j/i}$ de chi [pingguo]$_i$.
[they-PL]$_j$ [one-Cl following one-Cl]$_{j/i}$ DE eat [apples]$_i$.

(i) 'Every single person of them all eat one apple after another.' [SOA-OT]
(ii) 'Every single person of them in turn, all eat apples .' [SOA-ST]
(iii) 'They (possibly together) eat one apple after another.' [MA-OT]
(iv) 'They, one person after another, eat apples in turn.' [MA-ST]

TOO Annnnnnnoying!!! But!
Things become interesting when we add another element...

a. [tamen]$_j$ bixu/meiyou [**yige jie yige**]$_{j/i}$ de chi [pingguo]$_i$.
[they-PL]$_j$ must/NEG [one-Cl following one-Cl]$_{j/i}$ DE eat [apples]$_i$.
'[They]$_j$ must/didn't eat [apples]$_i$ [one by one]$_{j/i}$.' (Modal/Neg > *yige jie yige*: [MA-ST] √ [MA-OT] √)

b. [tamen]$_j$ [**yige jie yige**]$_{j/*i}$ de bixu/meiyou chi [pingguo]$_i$.
[they-PL]$_j$ [one-Cl following one-Cl]$_{j/*i}$ DE must/NEG eat [apples]$_i$.
'[They]$_j$ must/didn't eat [apples]$_i$ [one by one]$_{j/*i}$.' (*yige jie yige* > Modal/Neg: [SOA-ST] √, [SOA-OT]×)

Figure out the ambiguous syntactic structure: a **cartographic** point of view !

Control variables ST and OT for a more detailed result ——

Ensuring ST reading with an intransitive verb

a. tamen bixu yige jie yige de likai.
they must one-Cl following one-Cl DE leave
'They must leave in a manner of one by one.'
(Modal > *yige jie yige*, MA √, SOA ×)

b. tamen yige jie yige de bixu likai.
they one-Cl following one-Cl DE must leave
'One by one, they all must leave.'
(*yige jie yige* > Modal , MA ×, SOA √)

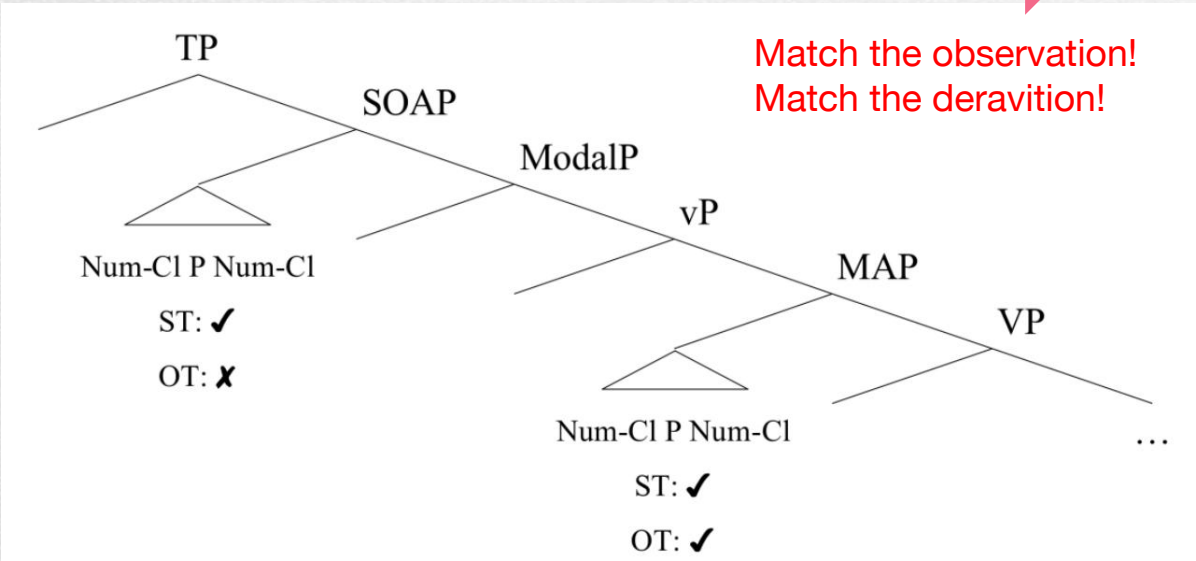
Ensuring OT reading with a singular subject

a. wo bixu yipian jie yipian de kan wenxian.
I must one-Cl following one-Cl DE read paper.
'I must read paper in a manner of one by one'
(Modal > *yipian jie yipian*, MA √, SOA ×)

b. * wo yipian jie yipian de bixu kan wenxian.
I one-Cl following one-Cl DE must read paper.
(*yipian jie yipian* > Modal , ungrammatical!)

The surface position

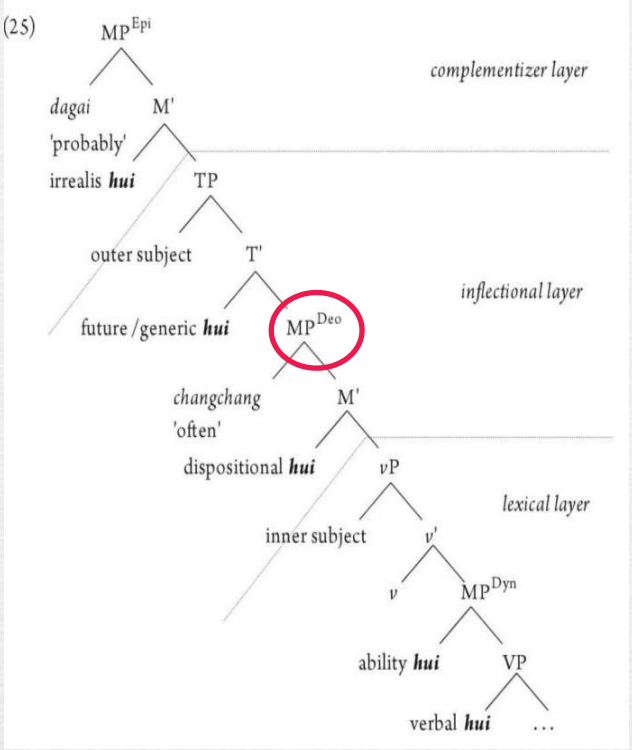
Under quantificational scopal interaction:



Match the observation!
Match the deravition!

Above ModalP: Subject-Oriented Adv.
Below ModalP: Manner Adv.

Deontic Modal



(Tsai, 2015)

Manner AP as Low V-complement

Generic Interpretation from Middle Voice → Manner Adv.

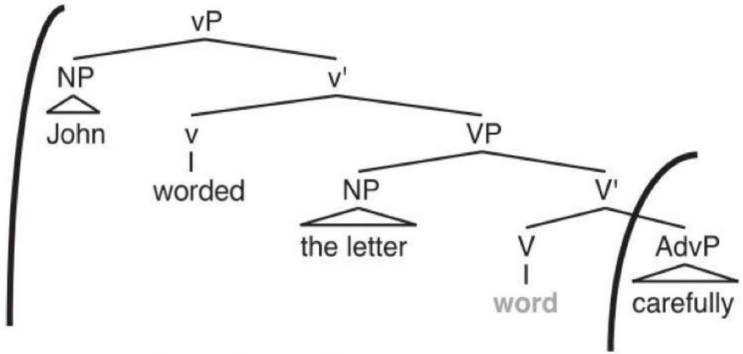
Mapping Hypothesis
(Diesing, 1992)

(25) John worded the letter carefully.

Γe[Con(e) & wording(o, e)] (careful(e))

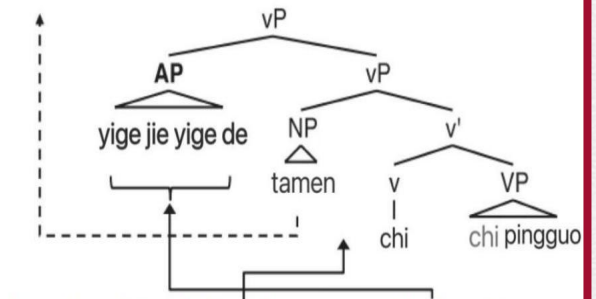
("Generally, for contextually relevant events involving John wording the letter, those wordings are careful.")

(26)



Quant	Restrictive Clause	Nuclear Scope
Γe	[Con(e) &wording(j,l,e)]	(careful(e))

However in Chinese...



(Stroik, 1990, 1992; Alexiadou, 1997; Larson, 1988, 2004, 2014, 2018)

AP-*DE* (地) as “concordializer” and move for Agreement

Agr. Domain: between a Case Probe and a Goal

Agreement-carrying inflections in European languages:

a. [$_{vP}$ v *küsste* [$_{DP}$ *das* *hübsche* *Mädchen*]]
kissed the.ACC pretty.ACC girl.ACC

b. [$_{vP}$ v … [$_{DP}$ D AP NP]]
[iACC[1]] [ACC[1]] [ACC[1]] [ACCval[1]]
③ ② ①
PROBE and *AGREE*

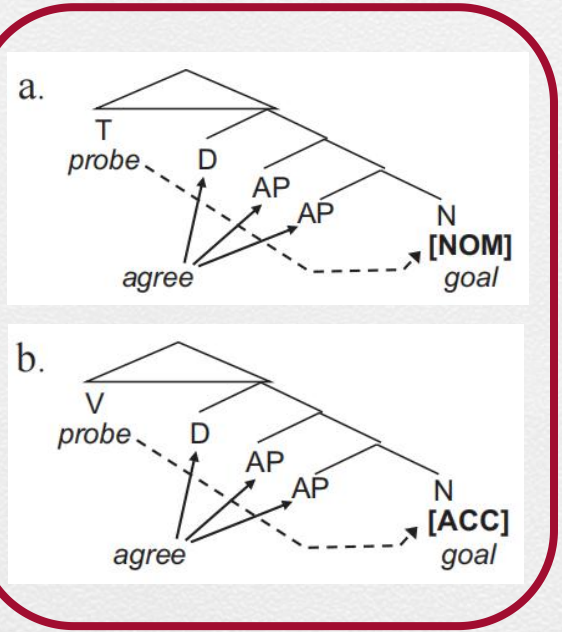
Application in Chinese:

[$_{T'}$ T [$_{vP}$ ta [$_{v'}$ v [$_{VP}$ jiang-hua [$_{NP}$ **dasheng-地**]]]]]
iNOM[1] NOMval[1] NOM[ ]
AGREE! *NO AGREE!* *ILLEGIBLE!*

[$_{T'}$ T [$_{XP}$ **dasheng-地**] [$_{vP}$ ta [$_{v'}$ v [$_{VP}$ jiang-hua dasheng-地]]]]]
iNOM[1] NOM[1] NOMval[1]
AGREE! *AGREE!* *LEGIBLE!*

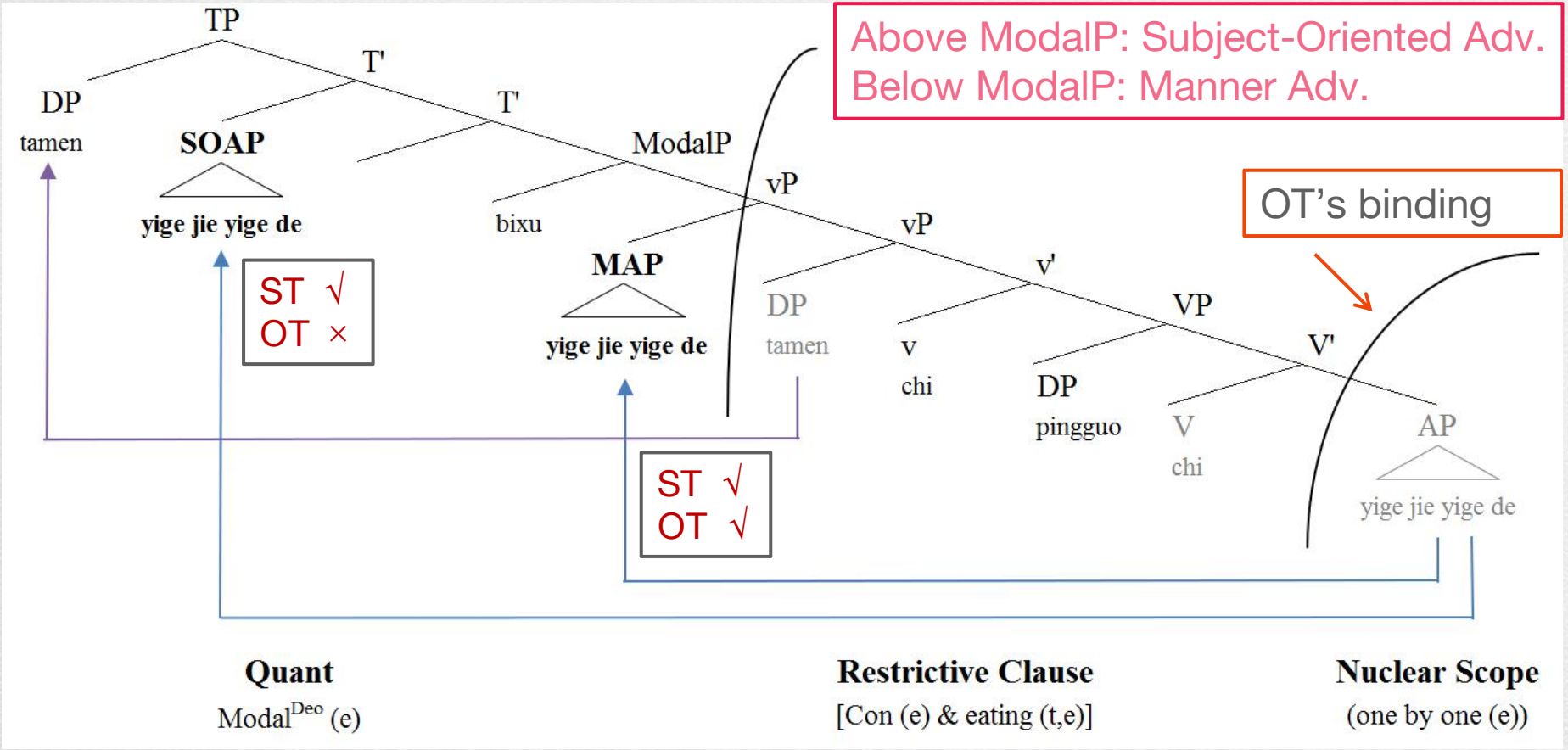
(大“声” - nominal)

Probe-Goal Approach



(Subject-in-vp hypothesis)

- Movement: *de* forces the NCPNC construction to move to its surface position to receive Case through Agreement



Reconstruction and Epiphenomenon

Reconstruction explains the Binding Relationship for OT reading:

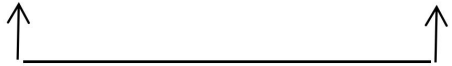
The Isomorphic Principle
(Aoun & Li, 1989)

Suppose A and B are quantifier phrases. Then if A c-commands B at S-Structure, A c-commands B at LF.

a. [$_{TP}$ tamen **[yige jie yige de]$_j$** [$_{VP}$ chi [pingguo]$_j$ **t$_j$**]]. (SOA)

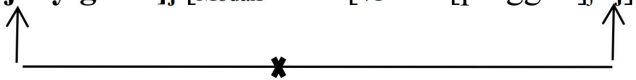


b. [$_{TP}$ tamen [$_{vP}$ **[yige jie yige de]$_j$** [$_{VP}$ chi [pingguo]$_j$ **t$_j$**]]]. (MA)

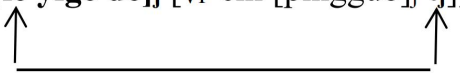


BUT with a quantificational element before, the reconstruction fails due to...

a. *[$_{TP}$ tamen **[yige jie yige de]$_j$** [$_{ModalP}$ bixu [$_{VP}$ chi [pingguo]$_j$ **t$_j$**]]. (SOA)



b. [$_{TP}$ tamen [$_{ModalP}$ bixu [$_{vP}$ **[yige jie yige de]$_j$** [$_{VP}$ chi [pingguo]$_j$ **t$_j$**]]]]. (MA)



Epiphenomenon of the ban against inverse scope in Mandarin !

05

Concluding Remarks

Therotical Implication
Further Study

Further Exploration

Scopal Relationship with Other Element

b. [tamen]$_j$ [yige jie yige]$_{j/*i}$ de qu chi [pingguo]$_i$.
[they-PL]$_j$ [one-Cl following one-Cl]$_{j/*i}$ DE go-for eat [apples]$_i$.
'[They]$_j$ go for eating [apples]$_i$ [one by one]$_{j/*i}$.'
(*yige jie yige* > Serial V, ST √, OT ×)

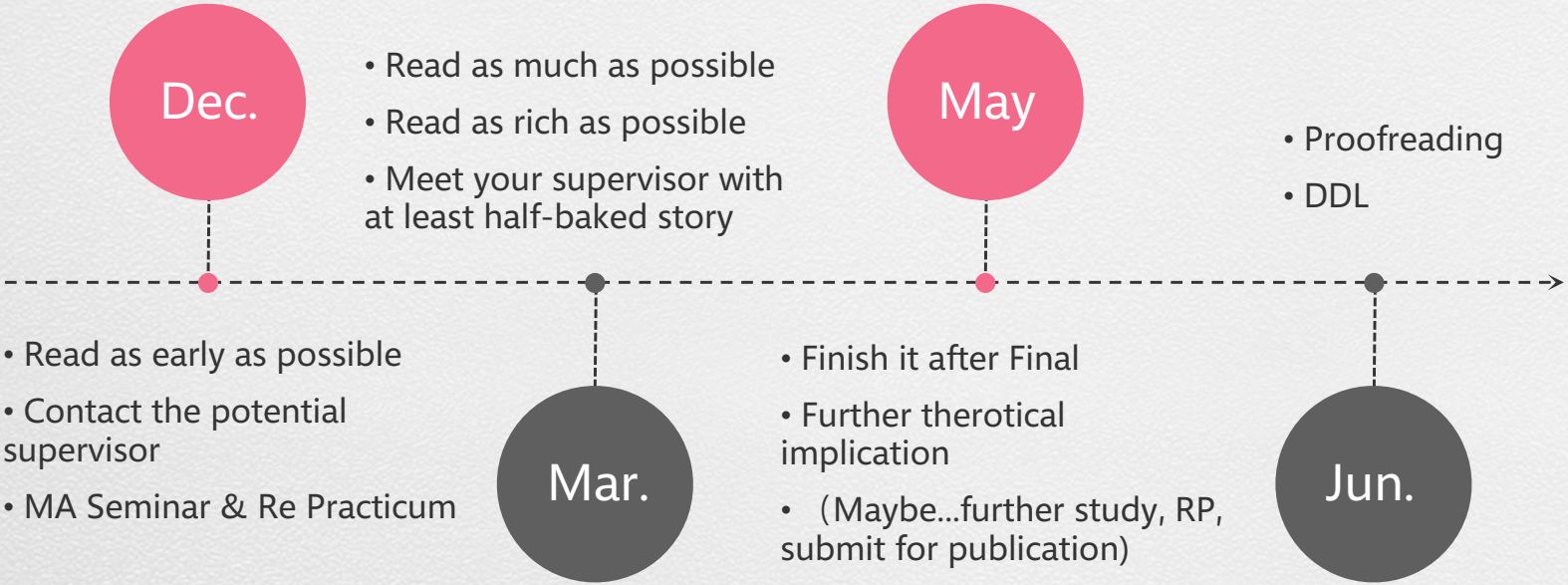
d. [tamen]$_j$ [yige jie yige]$_{j/*i}$ de ziji chi [pingguo]$_i$.
[they-PL]$_j$ [one-Cl following one-Cl]$_{j/*i}$ DE self eat [apples]$_i$.
'[They]$_j$ eat [apples]$_i$ [one by one]$_{j/*i}$ by themselves.'
(*yige jie yige* > Reflexive AP, ST √, OT ×)

The occurrence of *de* may affect the truth condition

a. [xueshengmen]$_j$ [yige jie yige]$_{j/i}$ de chi [pingguo]$_i$.
[student-Pl]$_j$ [one-Cl following one-Cl]$_{j/i}$ DE eat [apples]$_i$
'[Students]$_j$ eat [apples]$_i$ [one by one]$_{j/i}$.'

b. [xueshengmen]$_j$ [yige jie yige]$_{j/*i}$ chi [pingguo]$_i$.
[student-Pl]$_j$ [one-Cl following one-Cl]$_{j/*i}$ eat [apples]$_i$
'[One by one]$_{j/*i,}$ [students]$_j$ eat [apples]$_i$.'

Timeline & Tips





Thank You !

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